MATHS MANTHAN
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 05
Chapter 10: Circles

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A right triangle ABC with $\angle B = 90°$, $AB = 6$ cm, and $BC = 8$ cm circumscribes a circle. The inradius r of the circle is: [1]

(A) 1 cm (B) 2 cm (C) 3 cm (D) 4 cm

Q2. If PA and PB are two tangents drawn from an external point P to a circle of radius 4 cm such that $PA \perp PB$, then the length of chord AB is: [1]

(A) 4 cm (B) $4\sqrt{2}$ cm (C) 8 cm (D) $2\sqrt{2}$ cm

Q3. If radii of two concentric circles are 17 cm and 8 cm, the length of the chord of the larger circle tangent to the smaller circle is: [1]

(A) 15 cm (B) 25 cm (C) 30 cm (D) 32 cm

Q4. From an external point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is: [1]

(A) 7 cm (B) 12 cm (C) 15 cm (D) 24.5 cm

Q5. **Assertion (A):** If the angle between two tangents drawn from an external point P to a circle with centre O is 60°, then $\angle AOB = 120°$.
Reason (R): The angle between two tangents drawn from an external point is supplementary to the angle subtended by the line segments joining the points of contact at the centre. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. If PA and PB are tangents from external point P to a circle of radius 5 cm with centre O such that $PA \perp PB$, find the length of the chord AB. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. In $\triangle ABC$, an incircle touches sides BC, CA, AB at D, E, F respectively. If $AB = c, BC = a, CA = b$ and s is the semi-perimeter, prove that $AF = s - a$, $BD = s - b$, and $CE = s - c$. [3]

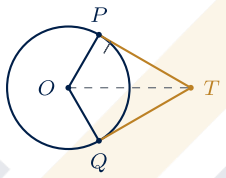
Q8. Tangents PA and PB are drawn to a circle with centre O from an external point P. If OP intersects the circle at Q and chord AB at M, prove that $PM \cdot PO = PA^2$. [3]

Q9. Two circles touch each other externally at point C. A common tangent AB touches the circles at A and B. Prove that the common tangent at C bisects AB, and that $\angle ACB = 90°$. [3]

SECTION D: Case-Based / Applied Competency Question (1 × 4 = 4 Marks)

Q10. Architectural Arch and Supporting Tangent Struts

An architect designs a semicircular decorative glass dome of radius $r = 4$ m resting on diameter AB with centre O. Two external support struts TP and TQ touch the circular boundary at points P and Q and meet at apex T. The struts are inclined to each other at $\angle PTQ = 90°$.



(i) Classify the geometric shape of quadrilateral $OPTQ$. [1]

(ii) Find the length of apex strut TP. [1]

(iii) Calculate the direct line-of-sight distance OT from centre to apex, OR find the length of chord PQ. [2]

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